

CSA0502 – QUERY PROCESSING FOR DATA SCIENCE

Lab Solutions – Set 2

Question 3

Write a Pandas program to display the details of jobs in descending sequence on job title.

Python Code:

```
import pandas as pd

# Create sample jobs dataset
jobs_data = {
    'JOB_ID': ['AD_PRES', 'AD_VP', 'AD_ASST', 'FI_MGR', 'FI_ACCOUNT', 'AC_MGR', 'SA_REP'],
    'JOB_TITLE': [
        'President',
        'Administration Vice President',
        'Administration Assistant',
        'Finance Manager',
        'Accountant',
        'Accounting Manager',
        'Sales Representative'
    ],
    'MIN_SALARY': [20080, 15000, 3000, 8200, 4200, 8200, 6000],
    'MAX_SALARY': [40000, 30000, 6000, 16000, 9000, 16000, 12000]
}

df = pd.DataFrame(jobs_data)

# Sort jobs details in descending sequence on JOB_TITLE
sorted_jobs = df.sort_values(by='JOB_TITLE', ascending=False)

print(sorted_jobs.to_string(index=False))
```

Output:

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
SA_REP	Sales Representative	6000	12000
AD_PRES	President	20080	40000
FI_MGR	Finance Manager	8200	16000
AD_VP	Administration Vice President	15000	30000
AD_ASST	Administration Assistant	3000	6000
AC_MGR	Accounting Manager	8200	16000
FI_ACCOUNT	Accountant	4200	9000

Question 4

Write a Pandas program to create a line plot of the historical stock prices of Alphabet Inc. between two specific dates.

Python Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Generate sample historical stock data for Alphabet Inc.
dates = pd.date_range(start='2023-01-01', end='2023-01-31')
np.random.seed(42)
close_prices = 88 + np.cumsum(np.random.randn(len(dates)) * 1.5)

df_stock = pd.DataFrame({
    'Date': dates,
    'Close': close_prices
})

# Filter data between two specific dates
start_date = '2023-01-05'
end_date = '2023-01-25'
mask = (df_stock['Date'] >= start_date) & (df_stock['Date'] <= end_date)
filtered_stock = df_stock.loc[mask]

# Create line plot
plt.figure(figsize=(8, 4))
plt.plot(filtered_stock['Date'], filtered_stock['Close'], marker='o', color='#1a73e8',
linewidth=2)
plt.title('Alphabet Inc. Historical Stock Prices (Jan 05, 2023 - Jan 25, 2023)')
plt.xlabel('Date')
plt.ylabel('Stock Price ($)')
plt.grid(True, linestyle='--', alpha=0.6)
plt.gcf().autofmt_xdate()
plt.tight_layout()
plt.show()
```

Output:

